# The assignment: this looks like a title

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|---|---|---|
| **Topic:** | Very Difficult Course (VDC) | |
| **Due:** | Jan. 01, 1970 by 23:59 hours | |
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| **Dated:** | Jan. 01, 2000 | |

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# 1 Introduction

This is an introduction chapter. We write these following similar syntax with `.md`.

```
= Chapter Title <ch:label>
== Section Title <sec:label>
=== Subsection Title <subsec:label>
```

## 1.1 Math

This is a section. Cross-referenced with Section 1.1 or the chapter with Section 1.

### 1.1.1 Inline mathematics

Inline math can be written as `$E = m c^2$`, rendered as: $E = mc^2$. Note that there is no space between `$` and math content. An inline integral is therefore `$integral_0^oo e^(-x^3) dif x$`. This is then rendered as $\int_0^\infty e^{-x^3}\,\mathrm{d}x$.

We also write greek letters and some math symbols with subscripts in similar fashion. We use `$alpha, beta, gamma$` for $\alpha, \beta, \gamma$ A subscript can be written with `$T_"system"$` for $T_{\text{system}}$.

### 1.1.2 Display equations

A numbered and labelled equation:

$$ax^2 + bx + c = 0 \tag{1}$$

An equation with cases:

$$f(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases} \tag{2}$$

A matrix equation:

$$\boldsymbol{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \tag{3}$$

### 1.1.3 Figures and tables

A figure placed at the bottom of a page, shown in Figure 1:

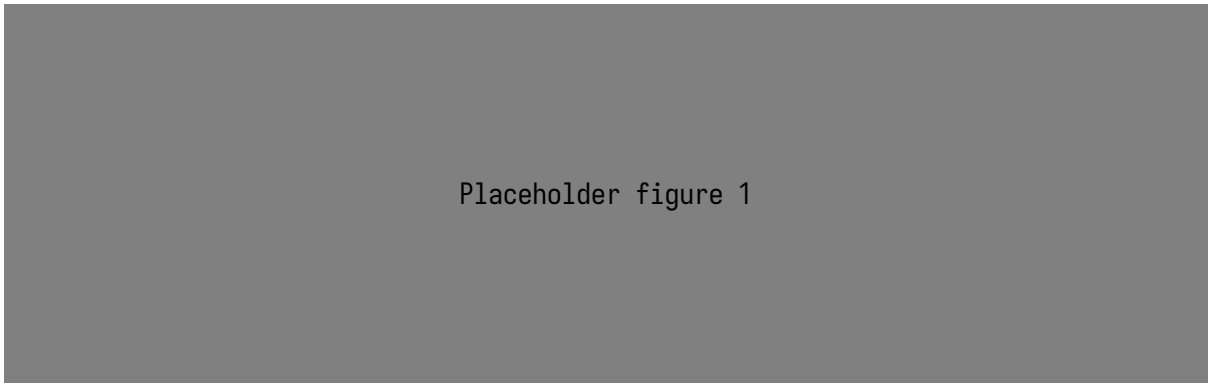
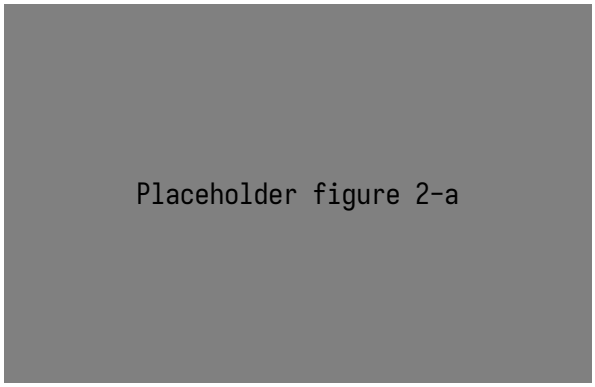



Figure 1: Example figure but with a longer caption.



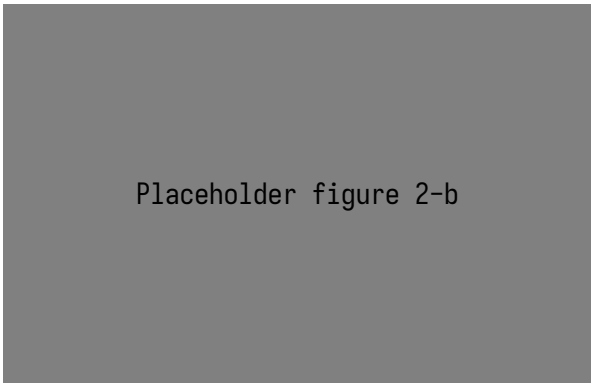




Figure 2: Two placeholder panels demonstrating side-by-side layout. (a) Left figure is much wow. (b) Right figure is also wow.

A side-by-side figure placed at top of the page is created via `grid` in Figure 2.

### 1.1.4 Code listings

Let us say that we want to include some python codes here. This can be achieved by following:

````
```python
import numpy as np

def simple_fn(a, b):
    """simply return (a + b)"""
    return a+b

```
````

This subsequently renders into:

```python
import numpy as np

def simple_fn(a, b):
    """simply return (a + b)"""
    return a+b
```

### 1.1.5 Citations

We can cite with keys showsn in a `.bib` file. The syntax is `@key`. A single entry with `item @key1` [1]. Multiple enteries as `item @key2 @key3` [2,3].

## 1.2 Problem and solution environments

**Problem 1.** What is the result of 1+1 in dec?
**Solution.** I believe this is 1+1=2.

## 1.3 Adding appendix

Appendix is initialized via `#show: appendix`.

## 1.4 Print bibliography

Print bibliography in a specific citation style from a `ref.bib` file with `#bibliography("ref.bib", style: "american-physics-society")`

## A Code listing

We list the codes from a code file:

```
"""
This is an example python code for Typst assognment tempalte.
"""

# import
import numpy as np

# func


def simple_fn(a: np.ndarray, b: np.ndarray) -> np.ndarray:
    """This is a simple function that goes a+b -> c"""
    return a + b


# main

if __name__ == "__main__":
    a = np.asarray([1, 2, 3])
    b = np.asarray([2, 4, 6])
    simple_fn(a, b)
```

## Bibliography

[1] W. I and M. This, *A Very Amazing Piece of Journal That You Should Read*, 3rd ed. (Fantastic publishing Co., Nightcity, 2077).

[2] H. May and I. Done, Yet another news for you, Journal of New Research **42**, 123 (2049).

[3] Contributors, *Should I Deploy Today*, https://shouldideploy.today/.